

Tutorial 11 Questions (ELL101)

1. Draw the circuit diagram to implement the expression:

$$y = AC + B\bar{C} + \bar{A}BC$$

2. Using Boolean algebra techniques, simplify this expression

$$F = AB + A(B + C) + B(B + C)$$

3. Draw the circuit to implement the function

$$F = ((A.B)' + (A.C)')'$$

Simplify this function and hence redraw the reduced circuit using only one gate with 3 inputs.

4. Simplify these Boolean expressions:

a) $A.\overline{A+B}$

b) $AB + A\bar{B}$

c) $AC + ABC\bar{C} + A\bar{B}C$

d) $ABC'D' + ABC'D + AB'C'D + ABCD + AB'CD + ABCD' + AB'CD'$

5. Form a Half-adder circuit with only NAND gate.

6. Given the logic function

$$F = \overline{AB + \bar{A}\bar{B}} + \bar{A}B$$

- a) Assuming the complements ($\bar{A}$ and $\bar{B}$) are available as inputs. Simplify the function using DeMorgan's theorem and synthesis it using the basic gates i.e. AND, OR and NOT
- b) Assuming the complements are not available, simplify the function and synthesis it from five NAND gates.
7. An electric light is to be controlled by three switches. The light is to be ON whenever switches A and B are in the same position; when A and B are in different positions, the light is to be controlled by switch C.
- (a) Draw up a truth table for this situation.
- (b) Represent the light function f in terms of A, B and C
- (c) Simplify the function and design a practical switching circuit.

8. Consider the following four-variable Boolean function:

$$F(A, B, C, D) = \sum (0, 2, 3, 5, 7, 8, 9, 10, 11, 13, 15)$$

Find out their reduced sum of product (SOP) terms using K-map.

9. Simply the Boolean function :

$$F(w, x, y, z) = \sum (1, 3, 7, 11, 15) \text{ and}$$

which has don't care conditions $d(w, x, y, z) = \sum (0, 2, 5)$

10. Simply the Boolean function:

$$F(x, y, z) = \sum (2, 3, 4, 5)$$